Dataset Proposal – NY SPARCS Hospital Inpatient Discharges

Members: Jin fa Zheng, Andrea Brugnera

Dataset

Name / source	Hospital Inpatient Discharges (SPARCS De-Identified), New York State Department of Health, published on Health Data NY
Licence	Open data under the NYS DOH Data Use Policy: free reuse with attribution to NYS DOH
Link	https://health.data.ny.gov/stories/s/wvua-rr23
Size	One file per year since 2009, about 2 million stays each (2,103,433 in 2022), 33–34 columns: hospital, admission type, demographics, diagnosis and procedure groups, severity, insurance, length of stay, charges and costs
Personal data	None: de-identified at source (age grouped, 3-digit zip codes, no exact dates)

Use case and organisation

A multi-hospital health system in New York State. Its problem is bed capacity: stays that run longer than expected block beds, so emergency patients wait, planned surgery is cancelled and unpaid bed-days pile up. A common cause is discharge planning that starts too late (nursing-home place, home care, equipment).

Decision and who takes it

Decision: at admission, which patients get early discharge planning, and how many beds each ward needs over the coming days. Today this happens case by case, usually after a delay has started.

Who takes it: the Director of Patient Flow, with the Head of Case Management, who assigns case managers.

Model support: predicted length of stay and a "prolonged stay" flag, using only information known at admission.

Metric

Primary KPI: excess bed-days per 100 admissions (days beyond the typical stay for the diagnosis group). **Supporting:** average length of stay, share of prolonged stays, cost of excess days. Baselines measured per year in WP1. **Target:** length of stay, plus a constructed flag for stays above the 75th percentile of their diagnosis group and severity.

Why this dataset

- **Real, messy data:** every inpatient stay in New York State, with all columns stored as text.
- **Constructed target:** the prolonged-stay label must be built, and discharge-only columns (charges, costs, destination) excluded to avoid leakage.
- **Relevant here too:** Spain's hospital discharge dataset (CMBD) has a similar structure, so the approach transfers.
- **Personal interest:** We believe there’s a real value and opportunity in this dataset and this problem; plus personal interest in the healthcare system